

DNA Isolation

Overview

Isolate plasmid DNA from bacterial cultures for concentration measurement and downstream applications.

A. Materials

- a. Bacterial Culture
- b. Mini Plus Plasmid DNA Extraction Kit (Viogene)
- c. Centrifuge
- d. Pipette
- e. Biodrop machine

B. Procedure

1. Transfer 1.8 mL of overnight bacterial culture to 2 mL microcentrifuge tube.
2. Centrifuge at 16,000 x g for 1 minute.
3. Discard supernatants.
4. Repeat steps 1–3 until all culture medium has been processed and pellets collected.
5. Use Viogene mini plus kit for DNA isolation.
6. Add 200 μ L of MX1 buffer.
7. Resuspend pellets by pipetting or vortexing.
8. Add 250 μ L of MX2 buffer.
9. Gently mix the buffer well (do not vortex) and invert 8–10 times.
10. Incubate the mixture at room temperature for 2 minutes.
11. Add 350 μ L MX3 buffer.
12. Immediately mix and invert the solution.
13. Centrifuge at 16,000 x g for 5 minutes.
14. While centrifuging, assemble and label column collection tubes.
15. Transfer supernatant (~900 μ L) to column and centrifuge at 7,000 x g for 30 seconds.
16. Discard flow-through.
17. Add 500 μ L WN buffer and centrifuge at 7,000 x g for 30 seconds. Discard flow-through.
18. Add 700 μ L WS buffer and centrifuge at 7,000 x g for 30 seconds. Discard flow-through.
19. Centrifuge empty column at 10,000 x g for 3 minutes to remove residual ethanol.
20. Place column into new 1.5 mL microcentrifuge tube.
21. Add 50 μ L elution buffer onto the center of the membrane and incubate the column at room temperature for 3 minutes.
22. Centrifuge at 10,000 x g for 3 minutes.
23. Turn on Biodrops and correct the settings.
24. Clean micro-volume sample port with ddH₂O, EtOH, then ddH₂O.
25. Measure the elution buffer as blank.
26. Measure the concentration, A_{260/230}, A_{260/280} of the samples and record on the tubes and in the lab notebook.
27. Clean micro-volume port with ddH₂O, EtOH, then ddH₂O.

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